

The partial derivatives are easily evaluated. You should prove for yourself that the two equations of motion are

$$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 + \frac{GMm}{r^2} = 0, \quad (10.6)$$

and

$$\frac{d}{dt}(mr^2\dot{\theta}) = 0, \quad \text{or} \quad \frac{dl}{dt} = 0. \quad (10.7)$$

The second of these equations gives

$$\dot{\theta} = \frac{l}{mr^2}.$$

Using this expression, Equation (10.6) leads to

$$\ddot{r} - \frac{l^2}{m^2r^3} = -\frac{GM}{r^2}. \quad (10.8)$$

These are, of course, the same as the equations of motion obtained using Newton's second law, namely Equations (10.4) and (10.5). Note, however, how much easier it is to use the Lagrangian than Newton's second law.

EXERCISE 10.5

Carry out the steps to show that $T = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2$.

EXERCISE 10.6

Obtain Equations (10.6) and (10.7).

EXERCISE 10.7

Suppose the force between a particle of mass m and a fixed point is given by $\mathbf{F} = -kr\hat{\mathbf{r}}$, where k is a constant. Obtain the Lagrangian and the equations of motion. Is angular momentum conserved for this system? Answer: $m\ddot{r} - mr\dot{\theta}^2 + kr = 0$; $\frac{d}{dt}(mr^2\dot{\theta}) = 0$.

10.5 Energy and the Effective Potential Energy

We have (twice!) obtained the equation of motion for the radial coordinate – Equation (10.5) and Equation (10.8). You probably expect to proceed by integrating it to get the value of r as a function of time and initial conditions. Indeed, we shall do that in Section 10.6, but first let us consider what the conservation of energy principle tells us about our problem. Since gravity is a conservative force (and there are no other forces), the total mechanical energy of our system is constant. Thus,

$$\begin{aligned} E &= \text{constant} = T + V \\ &= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - G\frac{Mm}{r}. \end{aligned}$$

Page 247

But $l = mr^2\dot{\theta}$, so we can replace $\dot{\theta}$ by l/mr^2 and write

$$E = \frac{1}{2}m\dot{r}^2 + \frac{m}{2}r^2 \left(\frac{l^2}{m^2r^4} \right) - \frac{GMm}{r},$$

or

$$E = \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r} \quad (10.9)$$

The first term depends on the square of the radial speed ($\dot{r}^2$) and the other terms depend only on position, so the first term *looks like* kinetic energy and the remaining terms *look like* potential energy. So it seems reasonable to associate a “radial kinetic energy” with the term $\frac{1}{2}m\dot{r}^2$ and an “effective potential energy” V_{eff} with the remaining two terms. The effective potential energy V_{eff} is given by⁶

$$V_{eff} = \frac{l^2}{2mr^2} - \frac{GMm}{r} \quad (10.10)$$

Although V_{eff} looks and acts like a potential energy and is a function only of position, it is definitely *not* a potential energy because it actually contains a kinetic energy term, namely $l^2/(2mr^2) = \frac{1}{2}mr^2\dot{\theta}^2$.

It is instructive to draw an energy diagram in terms of the effective potential energy.⁷ Note that V_{eff} is the sum of two terms, one positive and the other negative. For $r \rightarrow \infty$, the negative term in V_{eff} is the dominant term because

$$\frac{1}{r} \Big|_{r \rightarrow \infty} > \frac{1}{r^2} \Big|_{r \rightarrow \infty}.$$

As $r \rightarrow 0$, the positive term dominates because

$$\frac{1}{r} \Big|_{r \rightarrow 0} < \frac{1}{r^2} \Big|_{r \rightarrow 0}.$$

Therefore the plot of V_{eff} vs. r must have the general shape shown in Figure 10.4. Study this figure carefully and convince yourself it is qualitatively correct, specifically that V_{eff} is positive as $r \rightarrow 0$ and

negative as $r \rightarrow \infty$. Notice particularly that the $l^2/2mr^2$ term bends much more sharply than $-GMm/r$. Also, don't forget that r can only take on positive values.

Figure 10.5 is also a plot of $V_{eff}(r)$ vs. r . In this plot the effective potential energy does not have quite the right shape because I drew it to make it easy for you to appreciate various aspects of the effective potential that are hard to see on a more accurate plot, such as Figure 10.4. In Figure 10.5 you see four possible values for the total energy, denoted E_0 , E_1 , E_2 , and E_3 . Consider first a particle with energy E_1 . From Equations (10.9) and (10.10), we can write

$$\frac{1}{2}m\dot{r}^2 = E - V_{eff}. \quad (10.11)$$

⁶ The terms “effective potential energy” and “effective potential” are used interchangeably, even though a “potential” is actually potential energy per unit mass. Some books use the term “fictitious potential.” An older term is “centrifugal potential.” In general, the effective potential energy is defined as the sum of $l^2/2mr^2$ and the potential energy. Thus for an electron orbiting about a proton, the effective potential would be

$$V_{eff} = \frac{l^2}{2mr^2} - \frac{e^2}{4\pi\epsilon_0 r}.$$

⁷ At this time you may wish to review the material on energy diagrams in Section 5.6.

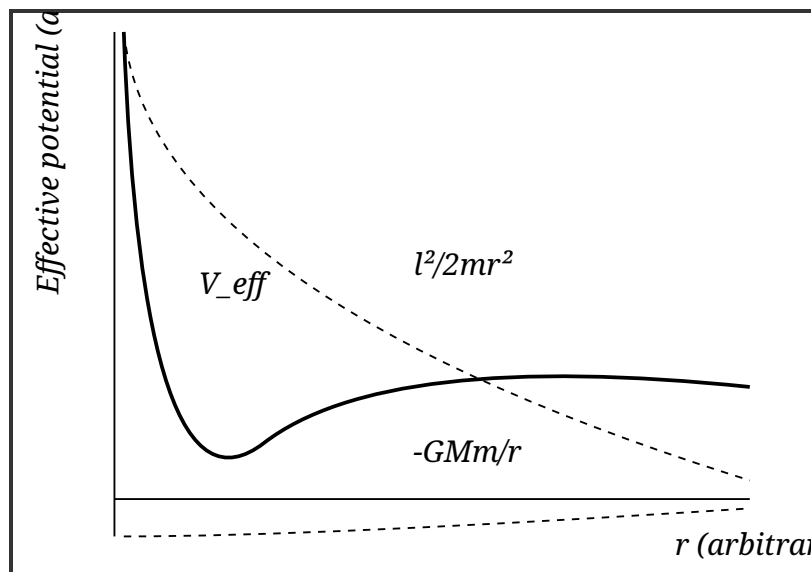


Figure 10.4 The effective potential energy $V_{eff}(r)$ is the sum of two terms, one positive and one negative.

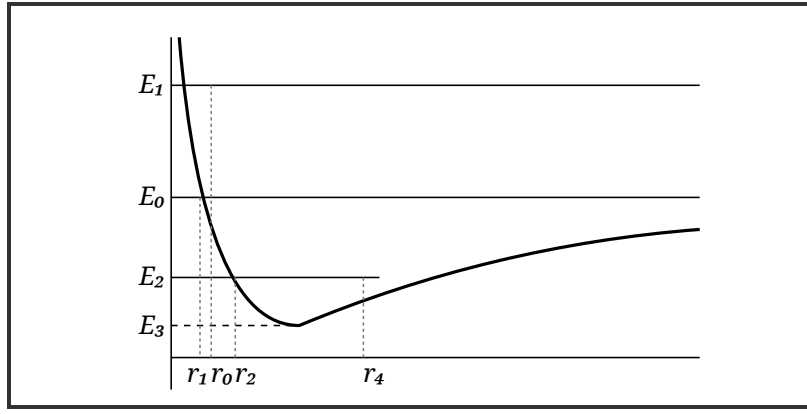


Figure 10.5 Energy diagram for the effective potential. The turning points for various values of total energy are indicated.

Because $\frac{1}{2}m\dot{r}^2$ can never be negative, the particle cannot be located at values of r for which $E < V_{eff}$. From Figure 10.5 this implies that if the energy is E_1 , then $r \geq r_1$, where r_1 is the point of closest approach. You can imagine the particle starting at $r = \infty$, coming closer and closer to the primary until it reaches r_1 (the turning point), and then moving back out to infinity. Note that according to Equation (10.11), the radial component of the velocity $\dot{r}$ is given by

$$\dot{r} = \sqrt{\frac{2}{m}[E - V_{eff}(r)]}. \quad (10.12)$$

Thus, the square of the radial speed is proportional to $E - V_{eff}$. (Note that $E - V_{eff}$ is not the kinetic energy, only the radial part of the total kinetic energy.) On the energy diagram (Figure 10.5) the distance between the horizontal line at E_1 and the heavy line representing V_{eff} is proportional to $\dot{r}^2$. At r_1 the value of V_{eff} is E_1 , so $\dot{r} = 0$. That is, at the turning point the particle has zero radial velocity, which makes perfect sense. At the turning point, the angular component of the velocity is a maximum because $r\dot{\theta} = (r)(l/mr^2) = l/mr$ is greatest when r is smallest.

Page 249

Let me emphasize that this not a complete description of the motion; it is only a description of the *radial* motion. Meanwhile the particle is also moving in θ with a velocity given by $r\dot{\theta} = l/mr$. The angular velocity increases as the radial distance decreases, in agreement with Kepler's second law. As we shall see, for positive values of the energy, the path of the particle is a hyperbola as illustrated in Figure 10.6. This could be the path of a comet coming in from infinity, speeding up as it approaches the Sun, swooping around the Sun, and then moving out to infinity. (Here "infinity" is a place far from the Sun, such as the Oort Cloud from which many comets are believed to originate.)

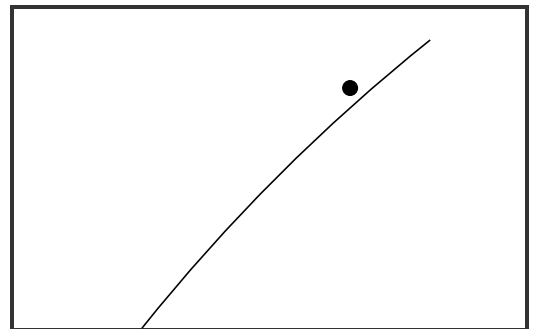


Figure 10.6 A comet or some other celestial object in a hyperbolic path.

Next consider a particle with zero total energy ($E = E_0 = 0$; see Figure 10.5). This means that the positive "radial kinetic energy" $\frac{1}{2}m\dot{r}^2$ is equal in magnitude to the negative effective potential energy V_{eff} . The motion of the particle as it comes from $r = \infty$ to the turning point at r_0 and then goes back out to $r = \infty$ is similar to the motion of the particle with energy E_1 . As we shall see, the main difference

is that the trajectory for energy $E_1 > 0$ is a hyperbola and the trajectory for energy $E_0 = 0$ is a parabola. For the parabolic orbit, the radial speed of the particle $\dot{r}$ is zero at infinity as well as at the turning point. As the particle comes in from infinity, $\dot{r}$ increases, reaching a maximum at r_4 where V_{eff} reaches its greatest negative value, then decreases to zero at r_0 . The angular velocity is given by $\dot{\theta} = l/mr^2$. (You should be able to describe the angular velocity as the particle comes in from infinity to the point of closest approach and then moves back out to infinity.)

If the particle has *negative* total energy, as indicated by E_2 in Figure 10.5, the motion is quite different; there are now *two* turning points and the particle can neither reach $r = 0$ nor move out to $r = \infty$. That is, the motion is *bounded*. The particle is trapped in a potential well. As it moves back and forth radially between the two points denoted by r_2 and r_3 , it is also moving azimuthally with a varying angular velocity $\dot{\theta}$. As we shall see shortly, this combination of radial and angular motion represents a trajectory which is an ellipse.

Finally, if the particle has energy E_3 , the minimum possible total energy, the value of $\dot{r}$ is zero at all times and the particle is at a constant radial position $r = r_4$. The path is a circle. The angular velocity is $\dot{\theta} = l/mr_4^2 = \text{constant}$. The particle is therefore moving with constant angular velocity in a circular path.

Depending on the value of the energy, the trajectory or orbit of the particle is a hyperbola, a parabola, an ellipse, or a circle. These are called conic sections because they can be generated by cutting a cone in various different ways, as shown in Figure 10.7.

We have been considering the motion of massive bodies interacting gravitationally, but the ideas and methods developed here are quite general and can be easily adapted to any central force problem, such as the motion of an electron in orbit around a proton in the Bohr model of the hydrogen atom.

EXERCISE 10.8

A particle is in a parabolic orbit. Where is its turning point? Answer: $r_0 = l^2/2GMm^2$.

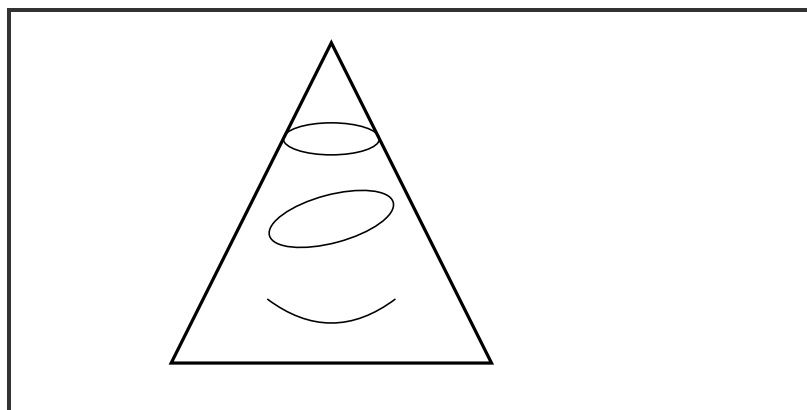


Figure 10.7 The conic sections. If the cone is cut as shown, the cross sections of the cuts will be a circle, an ellipse, a parabola, and one branch of a hyperbola.

EXERCISE 10.9

In the Bohr model of the hydrogen atom, an electron in the lowest orbit has angular momentum $l = \hbar$, where $\hbar$ is Planck's constant divided by 2π . The electrostatic potential energy of the electron and proton is $V = -e^2/4\pi\epsilon_0 r$. Show that the total energy (mechanical plus electrostatic) of the lowest orbit is $-(1/2)me^4/(4\pi\epsilon_0\hbar)^2$. Assume a circular orbit.

10.6 Solving the Radial Equation of Motion

The radial equation of motion, Equation (10.5), is a second-order ordinary differential equation of the form

$$\frac{d^2 r}{dt^2} = \frac{l^2}{m^2 r^3} - \frac{GM}{r^2}.$$

In previous chapters we learned two different ways of solving such an equation. In Section 3.6 we integrated directly by setting $\frac{d^2 r}{dt^2} = v \frac{dv}{dr}$. In Section 5.7 we solved for the motion by using the energy integral in the form of Equation (5.15), namely,

$$\int_{x_0}^{x(t)} \frac{dx}{\sqrt{E - V(x)}} = \sqrt{\frac{2}{m}} t.$$

Applying the energy approach to the present problem, we see that Equation (10.9) leads to

$$\frac{dr}{dt} = \left[\frac{2}{m} \left(E - \frac{l^2}{2mr^2} + \frac{GMm}{r} \right) \right]^{\frac{1}{2}}. \quad (10.13)$$

This equation yields the following definite integral

$$\int_{r_0}^{r(t)} \frac{dr}{\sqrt{\frac{2}{m} \left(E - \frac{l^2}{2mr^2} + \frac{GMm}{r} \right)}} = \int_0^t dt = t.$$

Integrating gives

$$t = t(r, r_0, E, l),$$

which can, in principle, be inverted to yield

$$r = r(t, r_0, E, l).$$

In this solution, the total energy E and angular momentum l are arbitrary constants. Note the use of the term *arbitrary* constant. Such constants are arbitrary in the sense that the differential equation is

satisfied by the solution no matter what values the constants happen to have. But for a particular problem, these constants are anything but arbitrary! In simple kinematics problems the “arbitrary constants” are usually the initial position and initial velocity. In more complex problems such as the one we just solved, they tend to be other constants of the motion such as the energy and angular momentum.

EXERCISE 10.10

Obtain $t = t(r, r_0, E, l)$ for a parabolic orbit (for which $E = 0$). (You will have to look up the integral.) Note that it would be very tedious to invert the expression to obtain $r = r(t)$. Answer:

$$t = \frac{\sqrt{2m}}{3(GMm)^2} \left(GMmr + \frac{l^2}{m} \right) \sqrt{GMmr - \frac{l^2}{2m}} - \frac{\sqrt{2m}}{3(GMm)^2} \left(GMmr_0 + \frac{l^2}{m} \right) \sqrt{GMmr_0 - \frac{l^2}{2m}}$$

10.7 The Equation of the Orbit

Many physicists work in the space program. In a few years you may be involved in planning a Moon shot or a planetary probe to study the clouds of Titan. If that happens, you will need to know the position of the spacecraft as a function of time. You will have to calculate $\mathbf{r} = \mathbf{r}(t)$ and you could use the technique described in the previous section.⁸

On the other hand, if you wish to describe the *orbit* of a satellite, planet or comet, you are not interested in its position at a particular time. Rather, you want a description of the path followed by the celestial object. Mathematically, this is given by the *equation of the orbit*, $r = r(\theta)$. Once you have this equation, you can determine r for every value of θ and thus trace out the path.

I will first present a “brute force” method and then a “sophisticated technique” for obtaining the orbit.

Brute Force Method

To determine the equation of the orbit, $r = r(\theta)$, start with the expression for $\frac{dr}{dt}$ obtained using the energy equation, Equation (10.13). Using the chain rule,

$$\frac{dr}{dt} = \frac{dr}{d\theta} \frac{d\theta}{dt} = \frac{dr}{d\theta} \dot{\theta}.$$

But recall that the angular momentum is $l = mr^2\dot{\theta}$, so

$$\frac{dr}{dt} = \frac{l}{mr^2} \frac{dr}{d\theta}. \quad (10.14)$$

⁸ There are other ways to determine the position of a space probe. These are described in engineering books on “astronautics.”

Substituting into Equation (10.13) yields

$$\begin{aligned}\frac{dr}{d\theta} &= \frac{mr^2}{l} \left[\frac{2}{m} \left(E - \frac{l^2}{2mr^2} + \frac{GMm}{r} \right) \right]^{\frac{1}{2}} \\ &= \left[\frac{2mE}{l^2} r^4 - r^2 + \frac{2GMm^2}{l^2} r^3 \right]^{\frac{1}{2}} \\ &= [\alpha r^2 + \beta r^3 + \gamma r^4]^{\frac{1}{2}},\end{aligned}$$

where I used the Greek letters α, β , and γ to illustrate the form of this equation. Rewriting and integrating the last equation gives

$$\int_{r_0}^r \frac{dr}{r(\alpha + \beta r + \gamma r^2)^{1/2}} = \int_{\theta_0}^{\theta} d\theta. \quad (10.15)$$

This integral can be found in tables of integrals.

Thus, in principle, the problem of determining the equation of the orbit is solved. The only things left to do are: (1) integrate Equation (10.15), and (2) invert the result to obtain $r = r(\theta)$.

Sophisticated Technique

There is a different way to obtain the equation of the orbit that cleverly avoids evaluating the complicated integral in Equation (10.15) and then carrying out the inversion to obtain $r = r(\theta)$. Bear with me for a little while because the math is a bit involved (but not difficult).

We will begin with Equation (10.2), the radial equation of motion:

$$\ddot{r} - r\dot{\theta}^2 = -\frac{GM}{r^2}.$$

Using $l = mr^2\dot{\theta}$ to eliminate $\dot{\theta}$ gives Equation (10.8) which is repeated here for convenience:

$$\ddot{r} - \frac{l^2}{m^2 r^3} = -\frac{GM}{r^2}.$$

Now let us introduce a new variable, u , defined as the inverse of r . That is,

$$u = \frac{1}{r}.$$

Then $r = u^{-1}$ and $dr = -(1/u^2)du$. Therefore,

$$\frac{dr}{dt} = -\frac{1}{u^2} \frac{du}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} \frac{d\theta}{dt},$$

where the last step uses the chain rule and the fact that $u = u(\theta)$. So,

$$\frac{dr}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} \dot{\theta} = -r^2 \dot{\theta} \frac{du}{d\theta}.$$

But $r^2 \dot{\theta} = l/m$. Replacing,

$$\frac{dr}{dt} = -\frac{l}{m} \frac{du}{d\theta}.$$

Taking the derivative with respect to time again,

$$\begin{aligned} \frac{d}{dt} \left(\frac{dr}{dt} \right) &= \frac{d}{dt} \left(-\frac{l}{m} \frac{du}{d\theta} \right), \\ \ddot{r} &= -\frac{l}{m} \frac{d}{dt} \left(\frac{du}{d\theta} \right) = -\frac{l}{m} \frac{d\theta}{dt} \left(\frac{d}{d\theta} \frac{du}{d\theta} \right), \\ &= -\frac{l}{m} \dot{\theta} \frac{d^2 u}{d\theta^2} \left(\frac{l}{mr^2} \right) = -\frac{l^2}{m^2 r^2} \frac{d^2 u}{d\theta^2}. \end{aligned}$$

But $1/r^2 = u^2$ so

$$\ddot{r} = -\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2}.$$

Substituting this into Equation (10.8) and using $1/r^3 = u^3$ we have

$$-\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2} - \frac{l^2}{m^2} u^3 = -GMu^2,$$

or

$$\frac{d^2 u}{d\theta^2} + u = \frac{GMm^2}{l^2}. \quad (10.16)$$

This is a very interesting equation. It looks like the equation for simple harmonic motion except for the additional constant term on the right-hand side. It can be made to look *exactly* like the simple harmonic motion equation by defining a new variable

$$w = u - \frac{GMm^2}{l^2}.$$

Then, since the last term is a constant,

$$\frac{dw}{d\theta} = \frac{du}{d\theta},$$

and

$$\frac{d^2 w}{d\theta^2} = \frac{d^2 u}{d\theta^2}.$$

Equation (10.16) can now be written in the form of the SHM equation:

$$\frac{d^2 w}{d\theta^2} + w = 0.$$

As noted in Section 3.7, the solution is sinusoidal and we can write

$$w = A \cos(\theta - \theta_0),$$

where A and θ_0 are integration constants. Consequently,

$$u - \frac{GMm^2}{l^2} = A \cos(\theta - \theta_0).$$

Now u is just the inverse of r , so the result finally gives an equation for r in terms of θ :

$$r = \frac{1}{u} = \frac{1}{\frac{GMm^2}{l^2} + A \cos(\theta - \theta_0)}.$$

Page 254

Dividing top and bottom by GMm^2/l^2 puts this in a nicer form:

$$r = \frac{l^2/GMm^2}{1 + \frac{Al^2}{GMm^2} \cos(\theta - \theta_0)} \quad (10.17)$$

From a mathematical point of view, the problem is now solved because r has been expressed in terms of θ and other known quantities (such as the angular momentum) and two constants of integration, A and θ_0 .

Equation (10.17) describes all the possible orbits for the two-body problem, i.e., circles, ellipses, parabolas, and hyperbolas. Because elliptical motion is of particular interest, let us assume that the total energy E is negative. Then the motion is bounded and the particle (planet) oscillates radially between turning points r_2 and r_3 as illustrated in Figure 10.5 for energy E_2 .

Equation (10.17) contains two constants of integration, A and θ_0 . We now consider their physical significance, beginning with A . At the turning points, the effective potential energy is equal to the total energy, $V_{eff}^{TP} = E$. This fact allows us to evaluate r_2 and r_3 which we write, in general, as r_{tp} . From Equation (10.10) we have

$$V_{eff}^{TP} = \frac{l^2}{2mr_{tp}^2} - \frac{GMm}{r_{tp}} = E.$$

Therefore, in terms of u ,

$$\frac{l^2}{2m} u_{tp}^2 - GMm u_{tp} - E = 0.$$

The two solutions of this quadratic equation for u_{tp} are:

$$u_{\pm} = \frac{m}{l^2} \left(GMm \pm \sqrt{(GMm)^2 + \frac{4l^2 E}{2m}} \right). \quad (10.18)$$

(Note that $1/u_{\pm}$ are the r_2 and r_3 turning points of Figure 10.5.)

But we have seen that

$$u = A \cos(\theta - \theta_0) + \frac{GMm^2}{l^2}$$

The maximum and minimum values of this expression occur when $\cos(\theta - \theta_0) = \pm 1$. That is,

$$u_+ = A + \frac{GMm^2}{l^2},$$

and

$$u_- = -A + \frac{GMm^2}{l^2}.$$

Using u_+ and equating the expression above to the relation given by Equation (10.18) leads to

$$A + \frac{GMm^2}{l^2} = \frac{GMm^2}{l^2} + \frac{m}{l^2} \sqrt{(GMm)^2 + \frac{2l^2 E}{m}},$$

or

$$A = \left[\frac{(GMm)^2 m^2}{l^4} + \frac{2Em}{l^2} \right]^{\frac{1}{2}}. \quad (10.19)$$

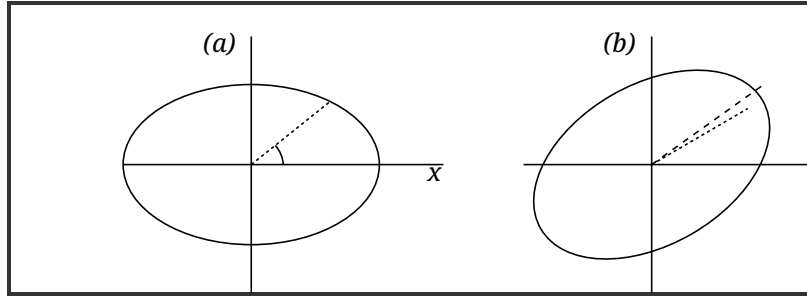


Figure 10.8 The position of a satellite relative to the major axis is given by the angle θ . (a) The major axis can be defined to be the x -axis. (b) The angle θ_0 gives the orientation of the elliptical orbit relative to a fixed line in space.

Thus A is related to the total energy and the angular momentum in a rather complicated way.

Let us now turn our attention to θ_0 , the other constant of integration. Considering Equation (10.17) we note that when $\cos(\theta - \theta_0) = +1$, the denominator is maximized and r reaches its minimum value. If M is the mass of the Sun, the place where r is a minimum is called the *perihelion*. The x -axis is usually taken to be a line from M through the perihelion, as shown in Figure 10.8. If θ is measured from the x -axis, then $\theta_0 = 0$.

On the other hand, if we measure θ from a fixed direction in space, such as the “first point in Aries” (Υ) then θ_0 is the angle between Υ and the x -axis and gives the orientation of the ellipse relative to the fixed direction.

We have, then, determined A in terms of E and l and given the physical interpretation of θ_0 as a description of the orientation of the ellipse with respect to inertial axes. Now the problem is indeed fully solved. Unfortunately, as you have no doubt noticed, our expressions are rather unwieldy. In a moment I will write them in a neater form. But first we need to consider some properties of conic sections, particularly ellipses.

10.8 The Equation of an Ellipse

You probably remember the grade school method for drawing an ellipse. You place a sheet of paper on a cork board and stick two tacks in it. Then you take a piece of string, tie the ends together, and loop it around the tacks. Finally you place a pencil in the loop of string and move it around the tacks, keeping the string taut, and if you are very careful you will draw an ellipse on the paper. See Figure 10.9.

As the pencil traces out the ellipse, the distance from the pencil to either tack varies. But the sum of the distances to the two tacks remains the same because the string is of constant length and the distance between the two tacks does not change. Therefore the distance from the first tack to the pencil *plus* the distance from the pencil to the second tack is constant. This fact is used for the mathematical definition of an ellipse.

An ellipse is the locus of points whose distances from two fixed points sum to a constant.

That is a bit complicated to say, but it is expressed quite simply as an equation. First choose the two fixed points, call them focal points and denote them F and F' . The distance from a point on the ellipse to F will be called r and the distance from that same point to F' will be r' . See Figure 10.10. Then an ellipse is defined as the locus of points such that $r + r' = \text{constant}$.

It is easy to obtain an equation for the ellipse in polar coordinates by introducing the angle θ between r and the major axis of the ellipse. The length of the semimajor axis ($\frac{1}{2}PP'$) will be...